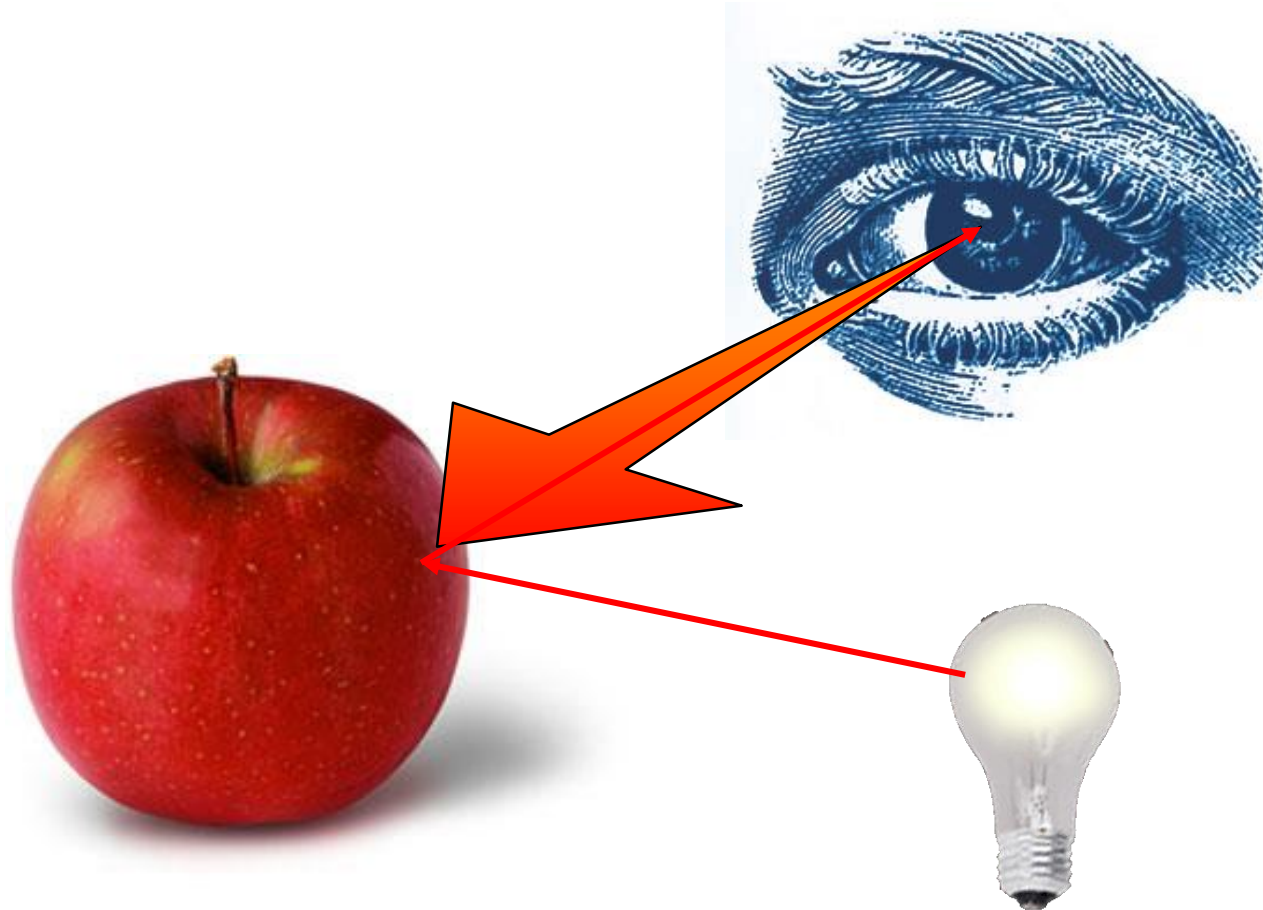


What is light?



Seeing the messengers?



Abu Ali Hasan Ibn Al-Haitham, ~1000 AD

He contradicted Ptolemy's and Euclid's theory of vision that objects are seen by rays of light emanating from the eyes; according to him the rays originate in the object of vision and not in the eye. Through these extensive researches on optics, he has been considered as **the father of modern optics**. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6074172/>

Particle theory of light

“The light and heat of the sun; these are composed of minute atoms which, when they are shoved off, lose no time in shooting right across the interspace of air in the direction imparted by the shove.”

Lucretius（卢克莱修），~50 BC

翻译：太阳的光与热，皆由微小的原子构成。这些原子一旦被弹射而出，便毫不耽搁，沿着推力赋予的方向，飞速穿越广阔的气隙。

作者与时代：卢克莱修（约公元前99年—约公元前55年），古罗马哲学家、诗人。这句话出自他的长诗《物性论》（De Rerum Natura）。这是现存最系统阐述古希腊原子论（由德谟克利特创立，伊壁鸠鲁继承）的拉丁文著作。

哲学内涵：

原子论：卢克莱修认为，世界万物（包括灵魂、光和热）都是由不可分割的“原子”在虚空中运动、碰撞、结合而成的。这与现代科学的物质观在精神上惊人地一致。

光的微粒说：这句话明确提出，光和热是由微小的原子（微粒）构成的，并且它们从光源（太阳）“弹射”出来，沿直线飞速穿越空间。这可以说是光的粒子性最早的诗意表达，比牛顿的微粒说早了1600多年。

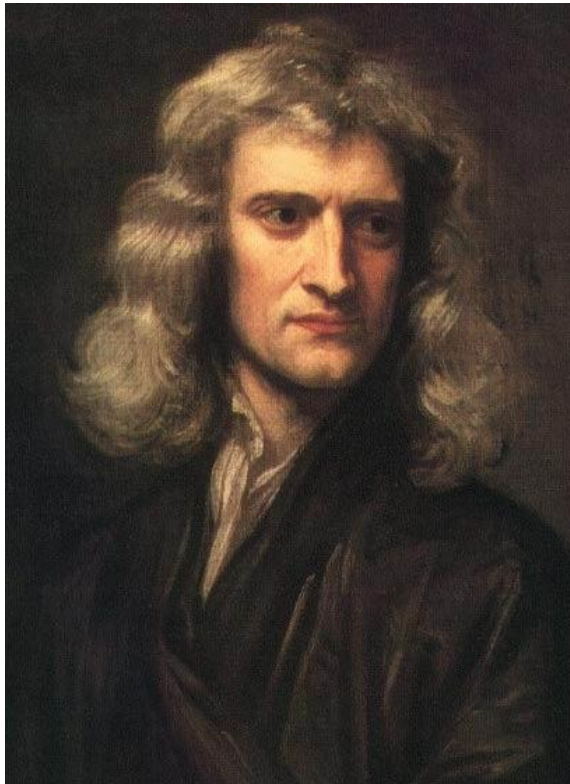
与现代科学的呼应：

在卢克莱修的时代，这完全是天才的哲学思辨。他认为光需要时间传播（“毫不耽搁”意味着有速度，而非瞬时到达），光沿直线传播（“沿着推力赋予的方向”），这些都与现代光学相符。当然，我们现在知道，光不是经典意义上的“原子”，但光子作为能量子，确实具有粒子性。

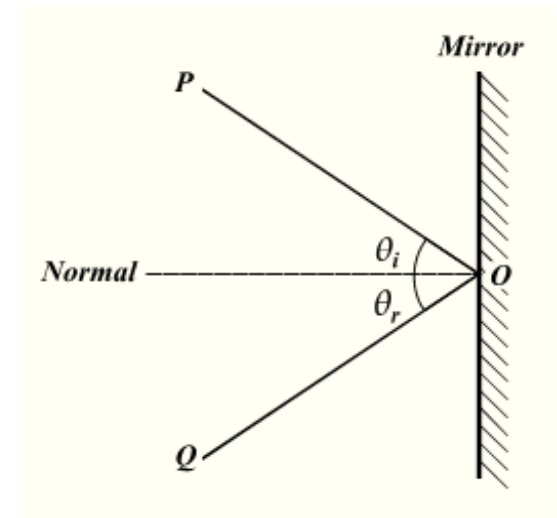
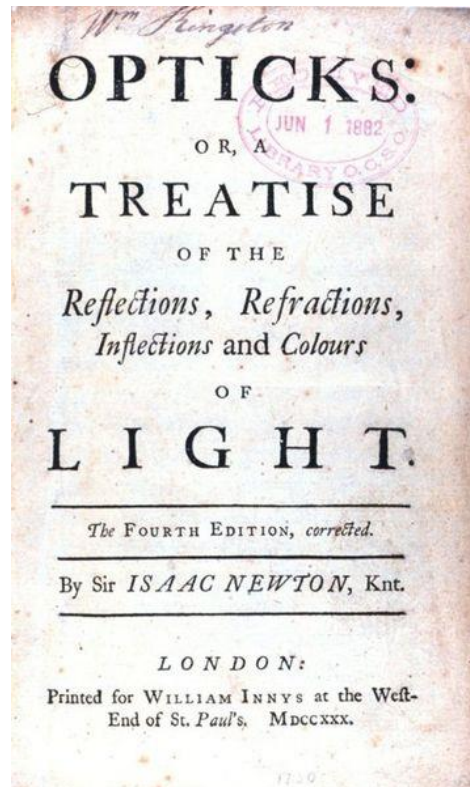
Particle theory of light

卢克莱修用优美的诗歌，在两千多年前就描绘了一幅朴素而深刻的光学图景——光，是由某种微小的、快速的东西构成的。

Reflection



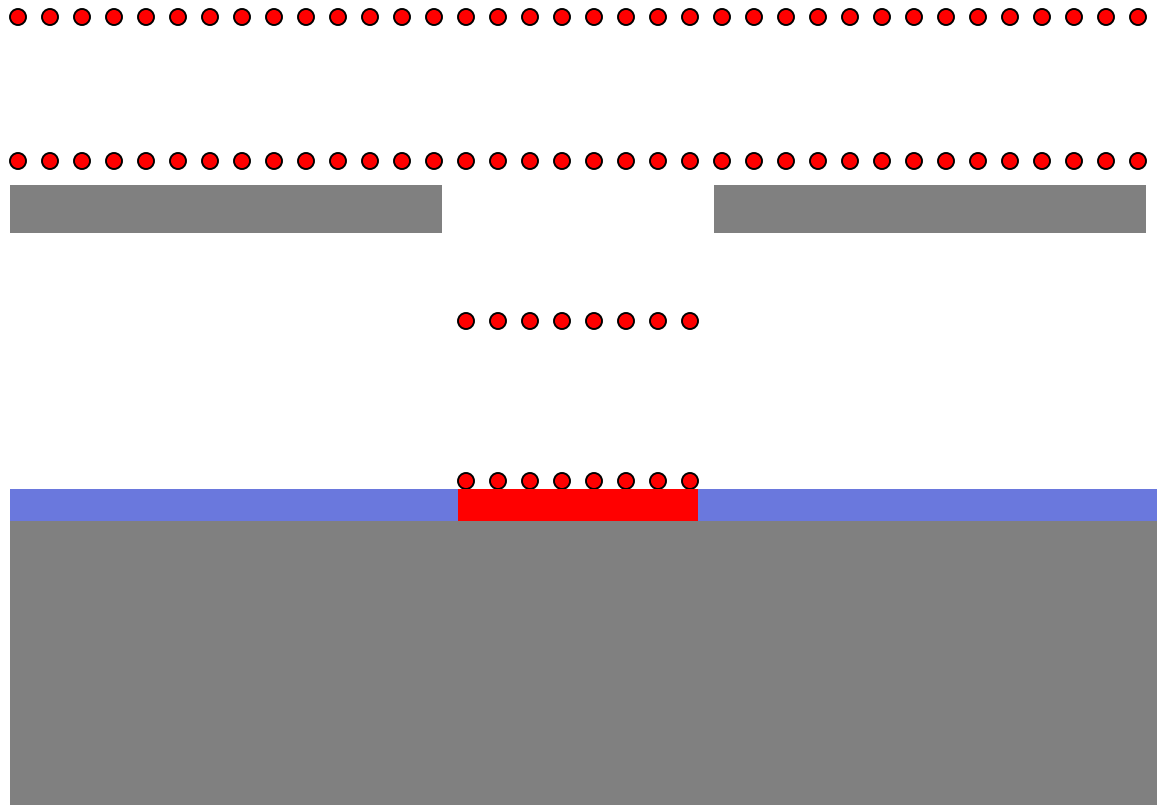
Sir Isaac Newton (1643-1727)



Billiards:

Particle theory of light

Light moves in straight lines



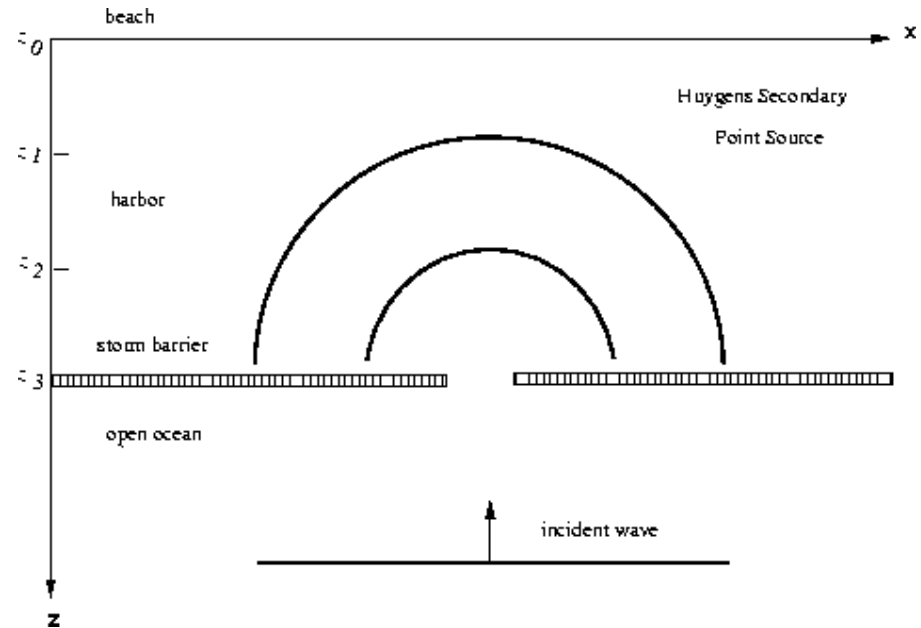
“Light does not bend into the shadow”

What will happen if the slit becomes smaller and smaller?

Wave theories of light

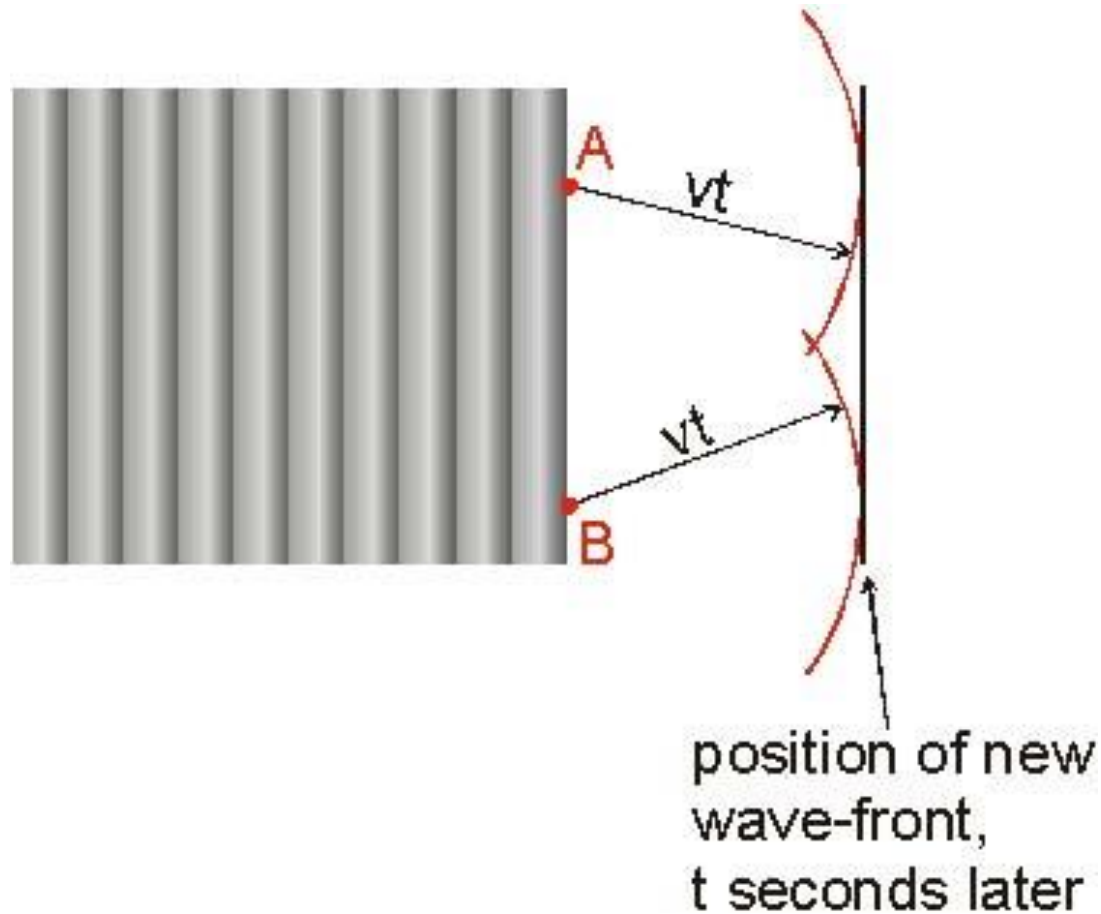


Christian Huygens (1629-1695)

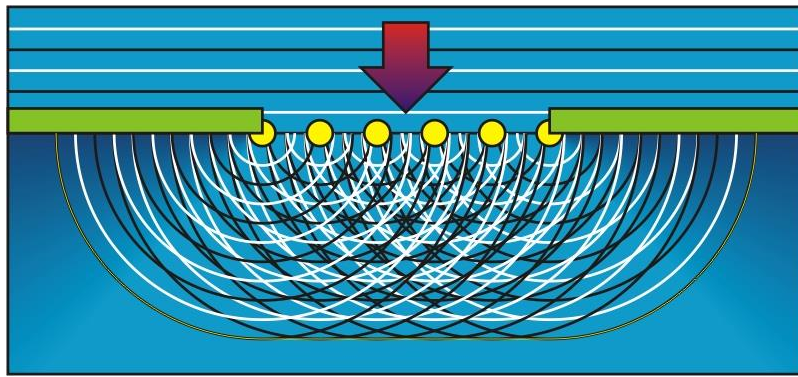


Each point on the existing wave-front can be considered to act as a source of waves (sometimes referred to as "secondary wavelets").

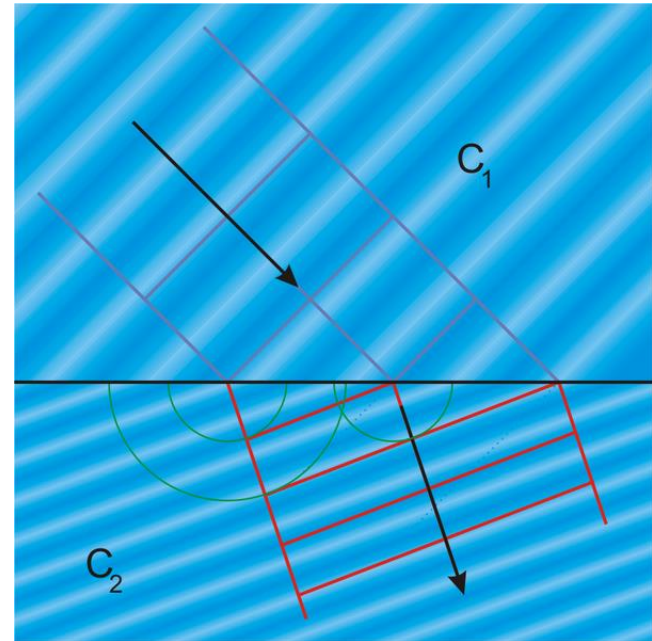
Wave theories of light



Wave theories of light



diffraction

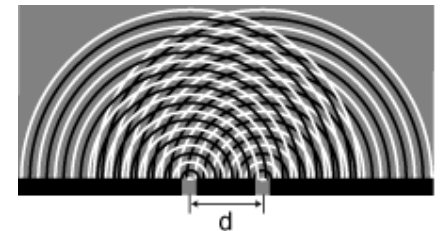
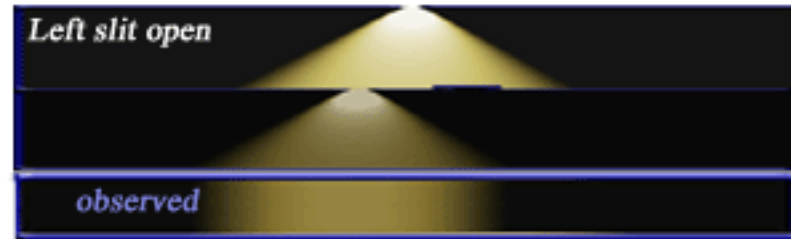


refraction

New observations



Thomas Young
(1773-1829)



Light is a wave!



Interference

Interference: Interaction of two or more light waves yielding a resultant irradiance that deviates from the sum of the component irradiances



If the high part of one wave (its crest) overlaps precisely with the high part of another wave, we get enhanced light.

Crest + Crest = Strong Light

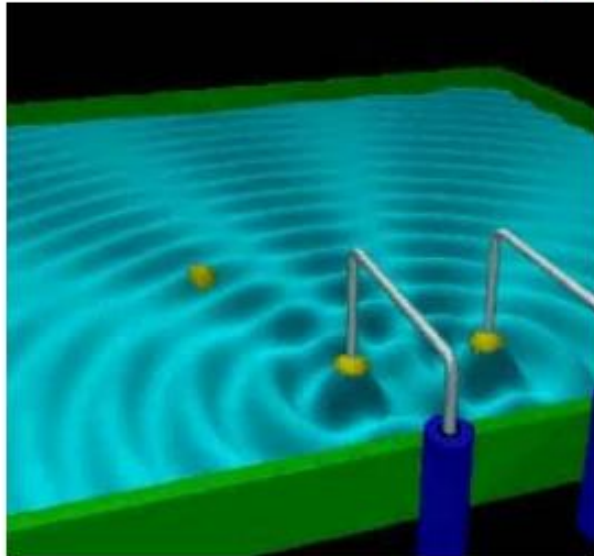
$$(r_1 - r_2) = 2\pi m / k = m\lambda$$



If the high part of one wave overlaps precisely with the low part of another wave (its trough), they cancel each other out.

Crest + Trough = Darkness

$$(r_1 - r_2) = \pi m / k = \frac{1}{2}m\lambda$$



Conditions of Interference:

- need not be in phase with each source, but the initial phase difference remains constant (coherent)
- A stable fringe pattern must have nearly the same frequency. But, white light will produce less sharp, observable interference
- should not be orthogonally polarized to each other

What kind of wave?

“This velocity is so nearly that of light, that it seems we have strong reason to conclude that light itself (including radiant heat, and other radiations if any) is an electromagnetic disturbance in the form of waves propagated through the electromagnetic field according to electromagnetic laws.”



James Clerk Maxwell

翻译：

这一速度与光速如此接近，以至于我们似乎有充分理由得出结论：光本身（包括辐射热，以及其他可能的辐射）就是一种电磁扰动，它以波的形式按照电磁定律穿过电磁场而传播。

背景说明：

这句话出自詹姆斯·克拉克·麦克斯韦。**这是物理学史上最伟大的宣言之一。**麦克斯韦在计算出电磁波的速度与当时已知的光速惊人地一致后，写下了这段话，从而正式宣告：光是一种电磁波。

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \cdot \mathbf{B} = 0$$

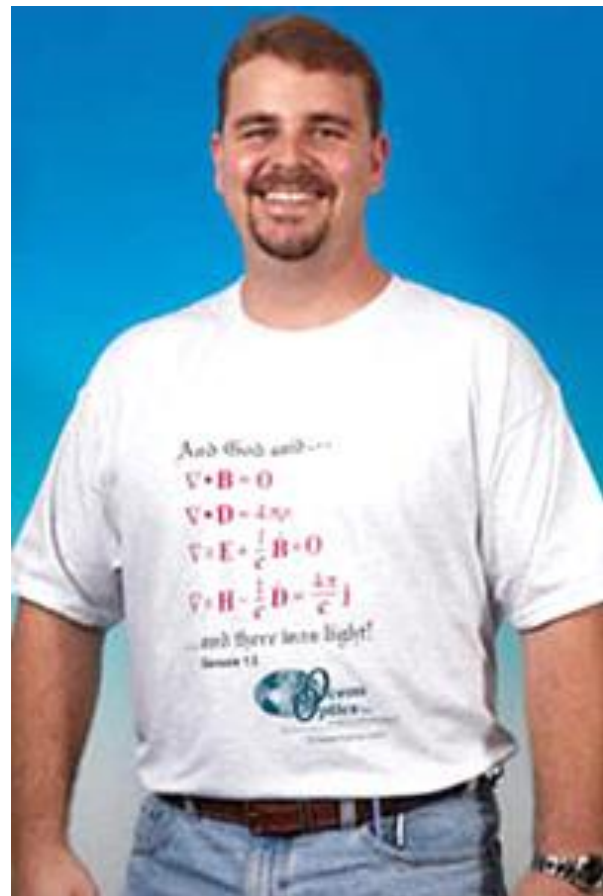
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

Crazy for Maxwell

为什么它很美？

- **对称性：** 变化的电场产生磁场，变化的磁场产生电场。在真空中，它们就像一对舞伴，相互扶持，彼此成就，共同前行。
- **自持性：** 一旦产生，不需要电荷或磁铁，它自己能靠相互感应在真空中无限传播下去。
- **速度：** 通过方程组推导出的波速表达式，代入数值后正好是光速。这不是巧合，这是光作为电磁波的铁证。



$$c = 1/\sqrt{\mu_0 \epsilon_0}$$

Experimental support

“在爱的鼓舞下他证实了电磁波的存在”

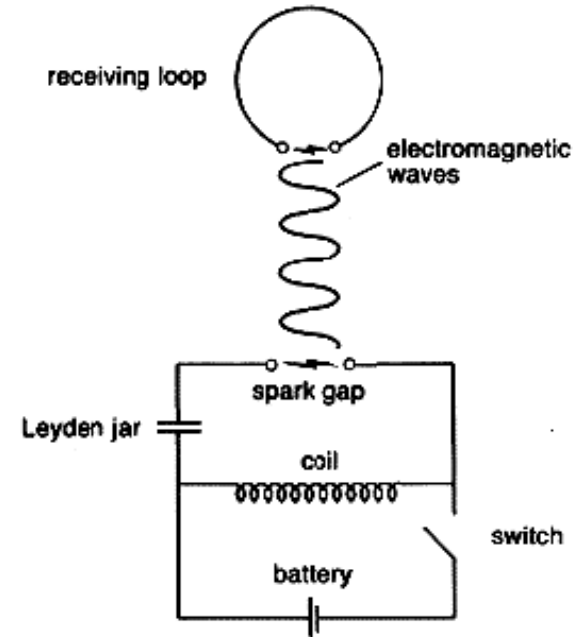
<https://baijiahao.baidu.com/s?id=1613907244945952809&wfr=spider&for=pc>

无数次失败并没有动摇赫兹的信心。他几乎是整日整夜地沉浸在实验之中。这期间他的艰苦可以从他写的一封信中看出：“**无论从时间上还是从性质上，我都象一个工人在工厂里那样工作，我上千次地重复每一个单调的动作，一个挨一个地钻孔、弯扁铁，接下来还要把它们涂上漆……**”



Heinrich Hertz
(1857 - 1894)

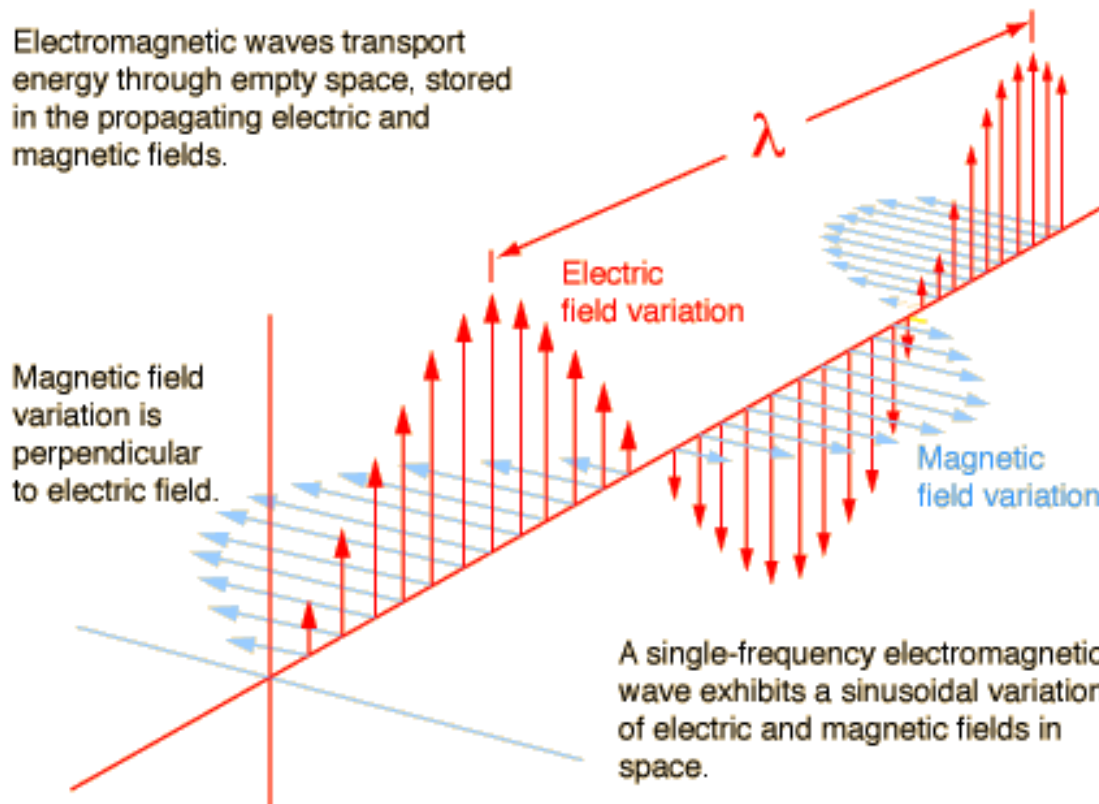
"I do not think that the wireless waves I have discovered will have any practical application."



1887年的一天，赫兹给振荡偶极子输入高压脉冲电流，象往常一样，在暗室中凝视共振偶极子铜环两个小铜球之间微小的空隙。突然，空隙中迸发出一个微弱的火花。他的助手高兴地喊出来：“有了，有火花了！”赫兹把铜环移到与发生器相距一定距离并适当地选择其方位时，他们看到电火花在两个铜球之间不断地跳跃。这样，赫兹在实验中初次观察到电磁振荡在空间传播。

Light as waves

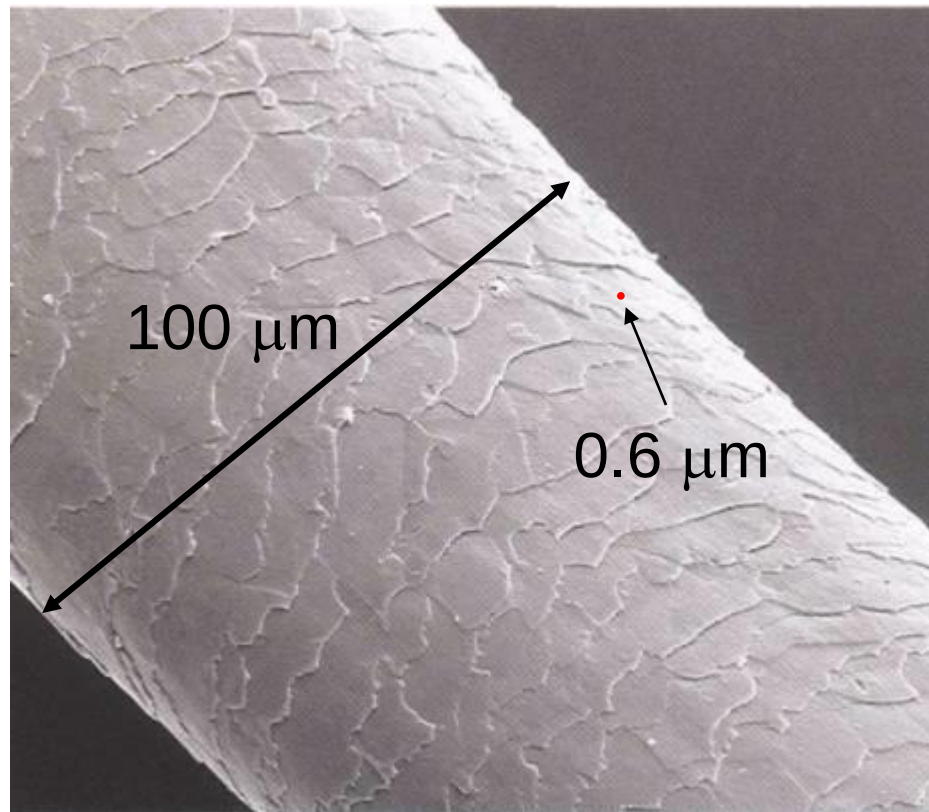
Electromagnetic waves transport energy through empty space, stored in the propagating electric and magnetic fields.



$$v = \frac{c}{\lambda}$$

Light as waves

■ $600 \text{ nm} = 0.6 \text{ }\mu\text{m}$



167 times as small as the diameter
of a hair!

Light as waves



Sodium vapor lamps are sometimes used for public lighting. They give off a yellowish light with wavelength of 589 nm. What is the frequency of this radiation?

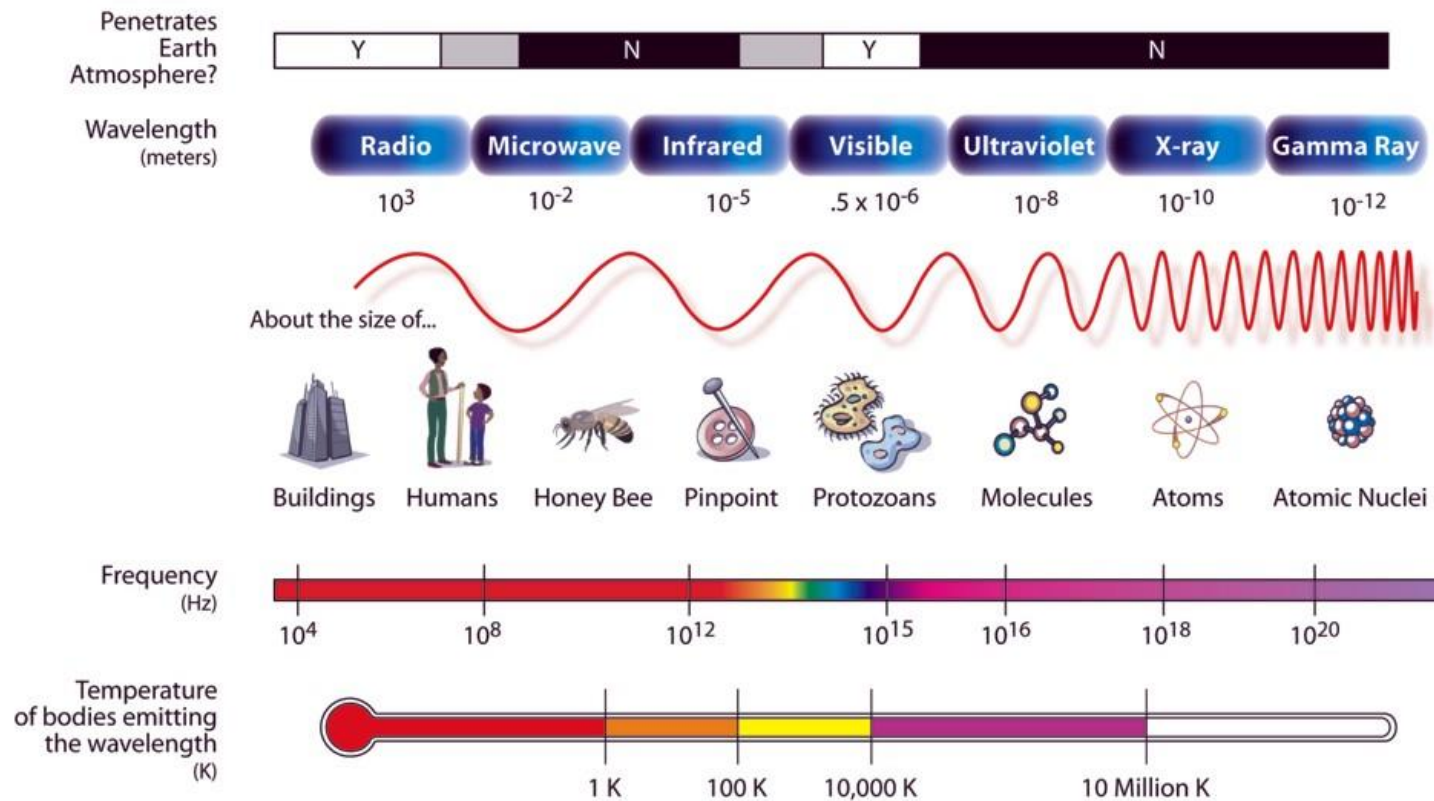
$$\nu = \frac{c}{\lambda}$$

$$= \frac{3.00 \cdot 10^8 \text{ m/s}}{589 \cdot 10^{-9} \text{ m}}$$

$$= 5.09 \cdot 10^{14} \text{ s}^{-1} = 5.09 \cdot 10^{14} \text{ Hz}$$

Spectrum of EM radiation

THE ELECTROMAGNETIC SPECTRUM



Are we done?

diffraction



衍射光栅:

<https://baike.baidu.com/item/%E8%A1%8D%E5%B0%84%E5%85%89%E6%A0%85/3173351?fr=aladdin>

refraction



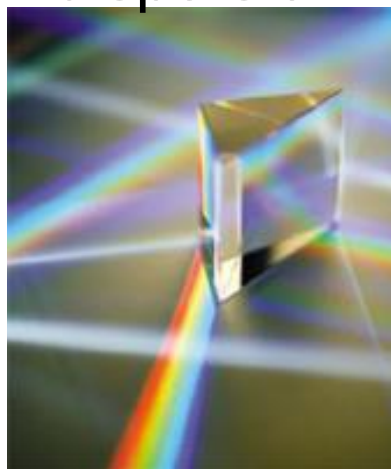
reflection



因为电子振动的幅度随频率变化，所以叠加后总的波速（或者说介质的折射率）也必然随频率变化。

波动说不仅能解释色散，而且正是通过解释色散，才确立了电磁理论的成功。海水的颜色、叶绿素为什么吸收红光和蓝光，都可以追溯到这些微观的共振模型。

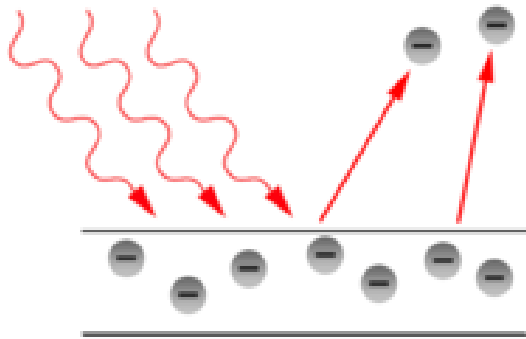
dispersion



反射定律不仅仅是一个几何事实，它是波在穿越不同介质时，为了保持自身一致性（**惠更斯原理**，在整个界面上所有点发出的次级波相干叠加），为了最快到达目的地（**费马原理**），最终由电磁场的基本规律（**麦克斯韦方程**）所决定的必然结果。

Particle nature of light

Photoelectric effect

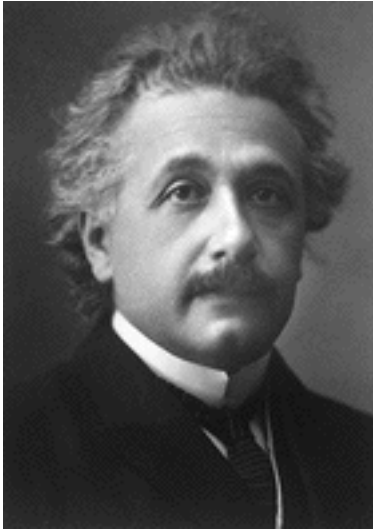


electrons are emitted:

- instantaneously when light is applied
- only when ν is higher than certain value
- below that ν , intensity doesn't matter

One-to-one interaction of a light quantum ($h\nu$) and the electron.

Atoms of light



“On a Heuristic Viewpoint Concerning the Production and Transformation of Light”, Einstein, 1905

《关于光的产生和转化的一个启发性观点》

- Light is composed of light quanta $E = h\nu$

历史地位：正是这篇论文提出的光量子解释，为他赢得了1921年的诺贝尔物理学奖（而不是相对论）。

这篇论文与卢克莱修的原子论、麦克斯韦的电磁波理论，构成了人类对光本质认识的三个里程碑。

- Light is electromagnetic radiation
“associated with singular points just like the occurrence of electrostatic fields according to the electron theory”

一个光量子（光子）是一个奇点（粒子性的能量包），它周围伴随着一个电磁场（波动性的场）。

“I therefore take the liberty of proposing for this hypothetical new atom, which is not light but plays an essential part in every process of radiation, the name **photon**”, Gilbert N. Lewis, 1926

Photons

Electromagnetic radiation has a frequency and a wavelength:

$$\nu = \frac{c}{\lambda}$$

Frequency is related to the energy of a photon:

$$E = h\nu = \frac{hc}{\lambda}$$

Photons are bosons ($s=1$), many photons can have the same 'state':

$$E_{\text{tot}} = nh\nu$$

Quantization of energy

A mole of photons in the yellow/red visible region of the spectrum has an energy of 195 kJ/mol. Calculate the wavelength of the light.

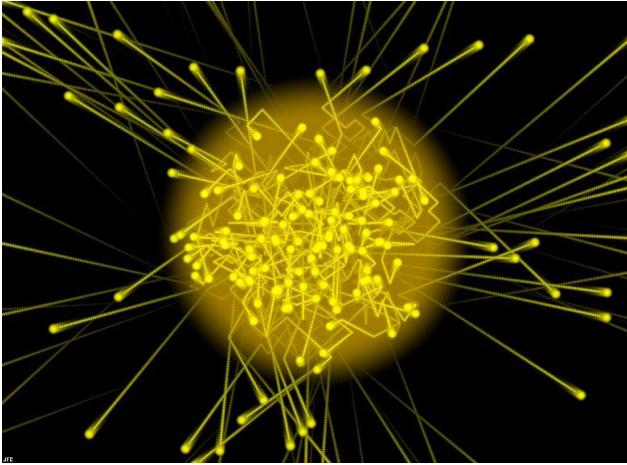
1. Calculate the energy for 1 photon.

$$\begin{aligned} E_{\text{photon}} &= \frac{195,000 J / \text{mol}}{6.022 \times 10^{23} \text{ photons} / \text{mol}} \\ &= 3.24 \times 10^{-19} J \end{aligned}$$

2. Find the wavelength.

$$\lambda = \frac{hc}{E} = \frac{(6.626 \times 10^{-34} Js)(3.00 \times 10^8 m/s)}{3.24 \times 10^{-19} J} = 6.14 \times 10^{-7} m$$

Do photons have mass?



Since photons have particle-like properties, they should have mass.

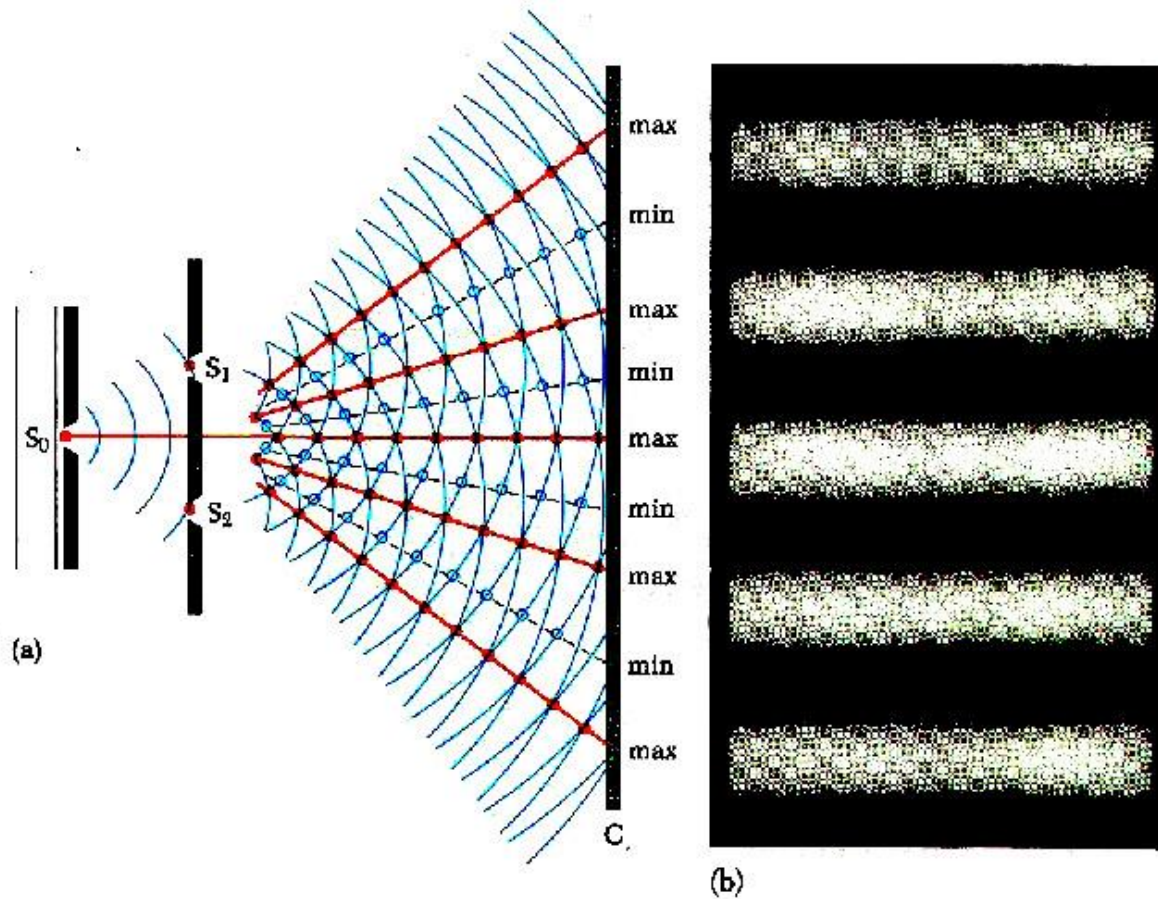
The (relativistic) mass of photons can be calculated from Einstein's equation for special relativity.

$$E = h\nu = \frac{hc}{\lambda}$$

$$E = mc^2$$

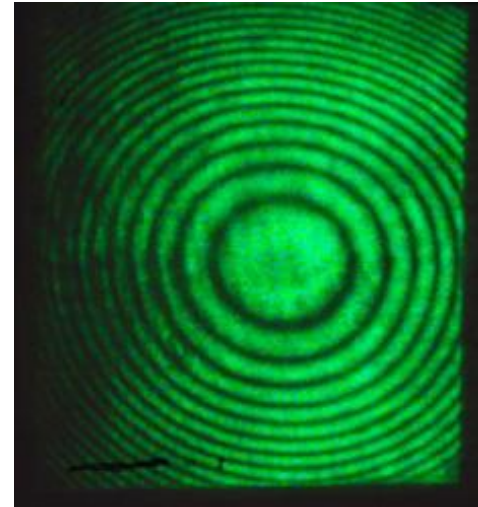
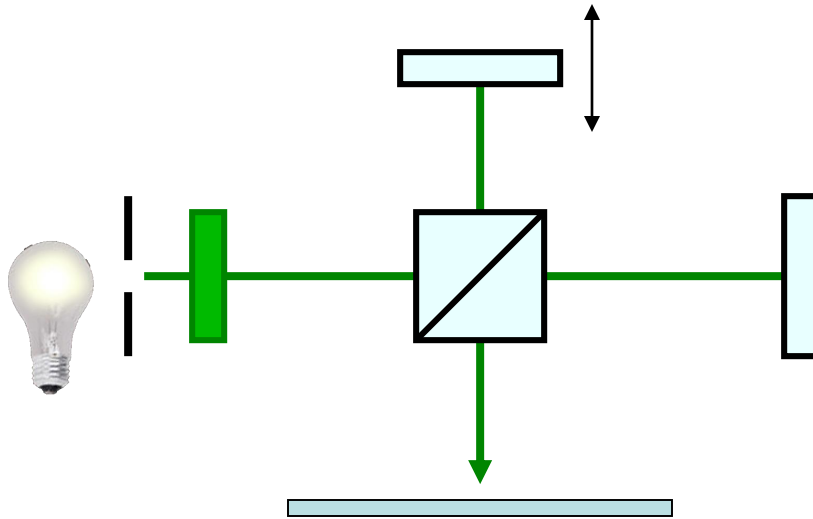
$$m = \frac{E}{c^2} = \frac{hc/\lambda}{c^2} = \frac{h/\lambda}{c}$$

Where's the photon?



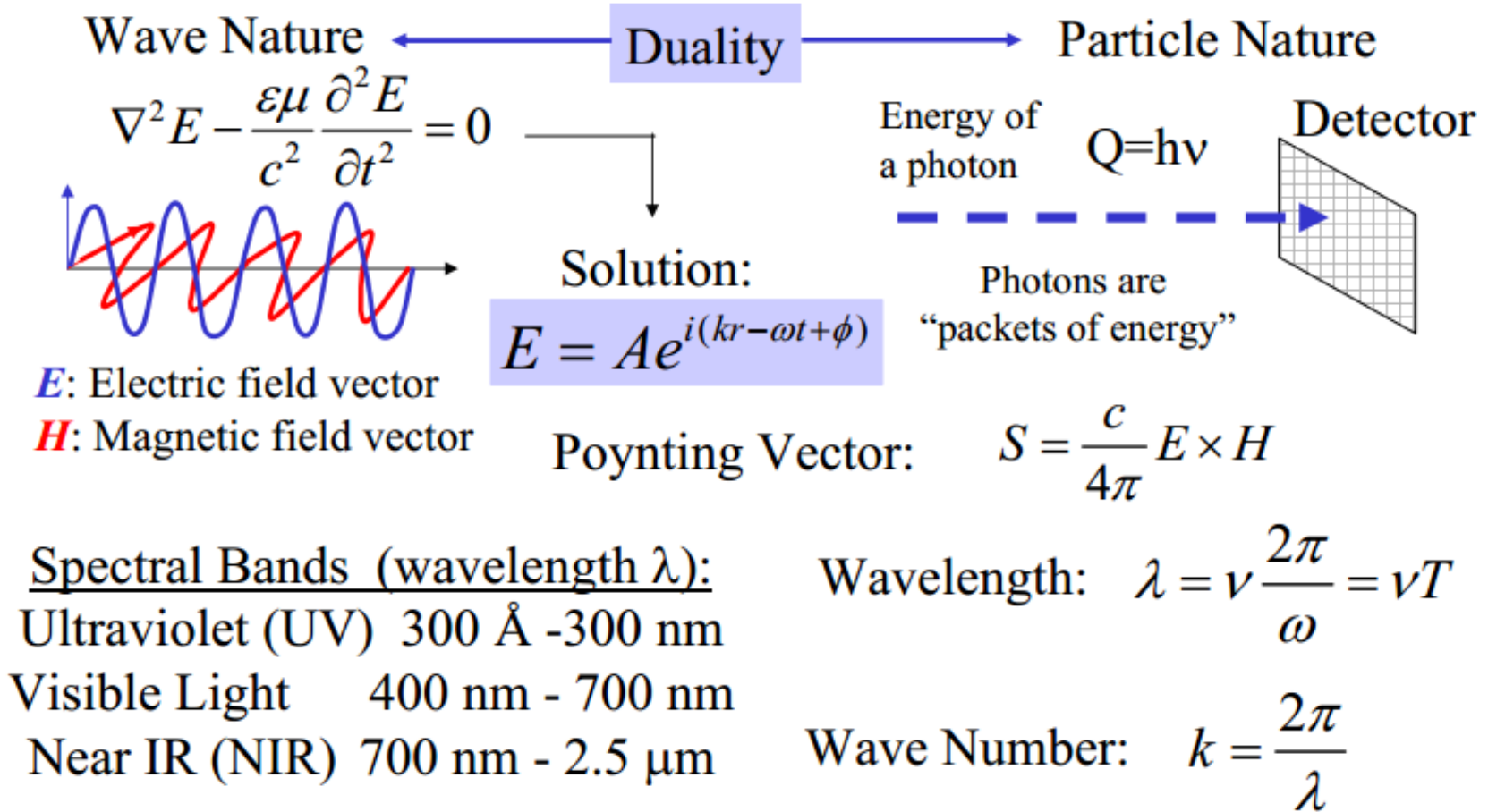
Where's the photon?

Single photon interference





Properties of Light



Strange particles

Photon 'state' is defined by:

- momentum
- Energy
- polarization
- spin ($s=1$)

Position and time (duration) of the photon are not well-defined!

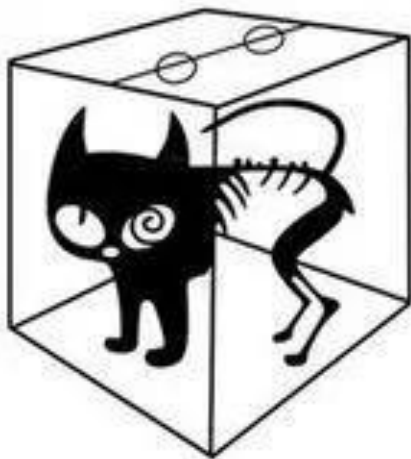
“Nobody knows, and it's best if you try not to think about it.”

Richard Feynman

“These days, every Tom, Dick and Harry thinks he knows what a photon is, but he is wrong”, Albert Einstein



SCHRODINGER'S CAT IS
A L E V E

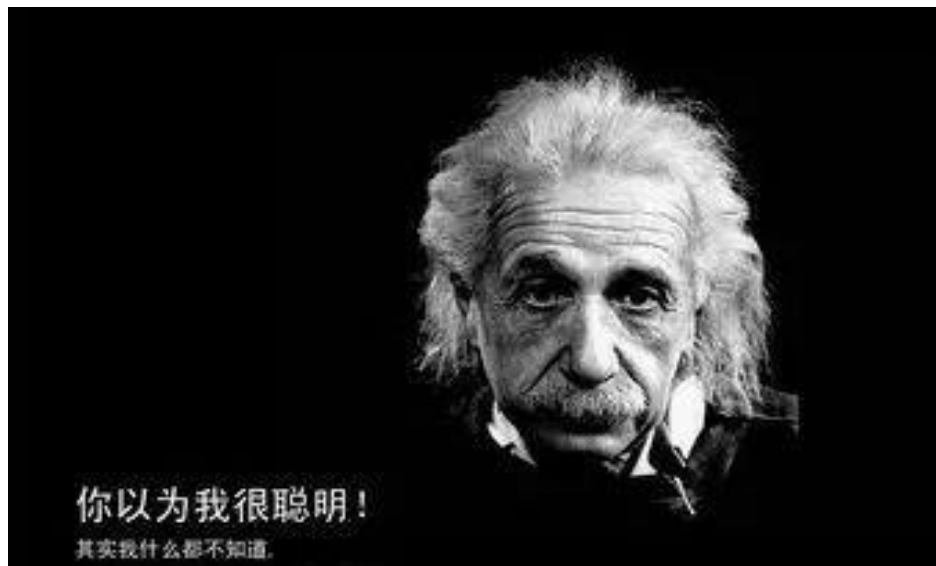


"I, at any rate, am convinced
that He [God] does not throw
dice."

~Albert Einstein
on Quantum Mechanics

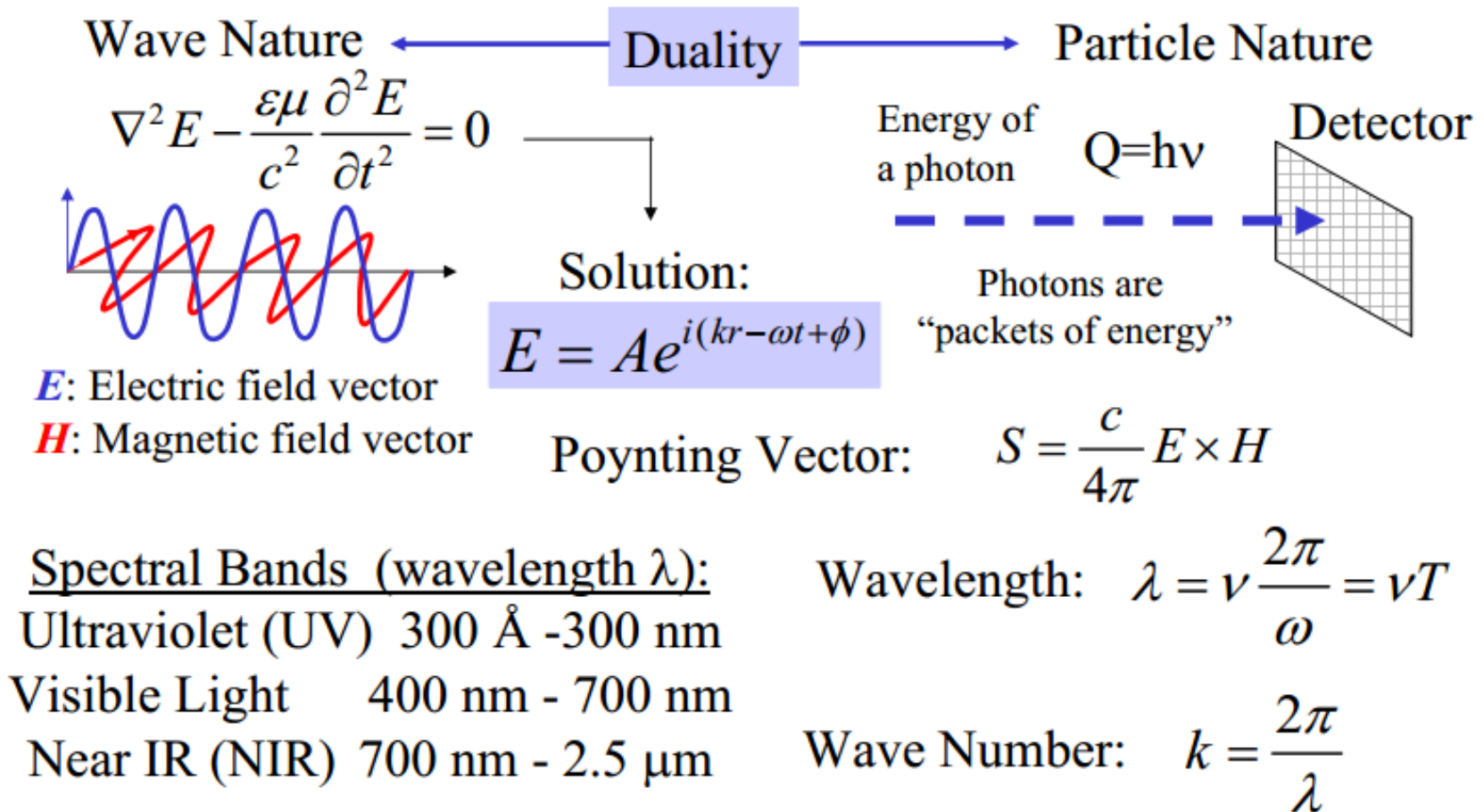
"Einstein, stop telling God
what to do."

~Niels Bohr on Einstein's
feeling about Quantum Physics





Properties of Light



More light on the horizon....

